As explained in the first chapter, one of virtual reality's pillars is immersion, or the extent to which the user is unaware of his actual physical surroundings. Virtual Reality is measured up against the actual Reality using two factors of immersion – the depth and the breadth. The Depth of the immersion refers to how complex the immersion is and how accurate and high quality the data received is. Display resolution, quality of graphics, audio output etc fall into this factor. The breadth of the immersion is defined as the number of senses involved in experiencing the reality.

Any sensory perception happens at a certain speed. When you turn your head, you do not expect your vision to follow later, even fractionally. When a train passes in front of you, you expect its sound to travel along as well, not behind it or not at all. When a virtual reality is being constructed, be it through software or hardware, the synchronization expected is the same. Any delay, called latency, is undesirable and reduces the degree of immersion.

Based on these ideas, there are quite a few types of VR products currently in the market as explored in the previous chapter. Let us have a deeper look at how each of them simulates their part of reality.

The Headset Display

Currently, there are two types of Head Mounted Displays in mainstream VR - ones that leave the computation and some of the tracking to external computers and connected devices, and ones into which you can put your smartphones and use it as a body that encases it. Both follow certain principles to generate the visual experience that they do.

Our sense of vision combines slightly different images from each eye to give us a sense of depth and perspective. To achieve this, the headset either displays two different images on one screen or uses two different screens for separate images. Between this display and the user's eyes, lenses are used to tilt and alter a 2D flat image to a 3D stereoscopic image in a way that our eyes perceive it the same as real life visuals. The distance between the two lenses can be altered in most headsets to adjust for the variable distance between the two eyes of different people. The illusion of depth thus achieved, is maintained even when the view is shifted.

This shifting of the view is possible due to a wide field of view. Although most devices nowadays are aiming for a 100 to 110 degree view, for a truly immersive and mobile VR experience, a 360 degrees display is not overkill. Along with that, a suitably high refresh rate, not less than 60fps, is necessary